Misallocation and Variable Markups

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November 5, 2026

Misallocation

- Why are there differences in GDP per capita across countries?
  - Many potential explanations: slow diffusion of technology, frictions in moving inputs across space, differences in factor endowments, etc.
- One theory that has recently gained attention is **misallocation**: low-income countries are not as effective in allocating factors of production to their most efficient use.
- Potential origins of misallocation: uneven tax regulation, corruption, market imperfections, trade barriers, ...

Misallocation: Intuition

- Idea: suppose firms produce in some Cobb-Douglas way:

$$y_i = A_i K_i^{\alpha} L_i^{1-\alpha}$$

- In a competitive market, we would expect the marginal product of labour to be equal to the wage, and the marginal product of capital to be equal to the rental rate:

$$\frac{\partial y_i}{\partial L_i} = w, \quad \frac{\partial y_i}{\partial K_i} = r$$

- Using more units of L_i or K_i than optimal is inefficient because of diminishing marginal products

Misallocation: Intuition cont.

- Take two firms with identical A_i, and suppose firm 1 has access to cheap interest rates because of government favouritism
  - Could be corruption, could be that firm 2 faces a higher capital tax, etc.
- If both firms equate marginal products to marginal costs, then firm 1 will use capital less efficiently than firm 2
- In other words, if we had taken some of firm 1's capital and given it to firm 2, total output would increase!
- To be determined: how large are the effects of misallocation, and to what extent can they explain cross-country GDP per capita differences?

Misallocation: Framework

- Model from Restuccia & Rogerson (2008) and Hsieh and Klenow (2009)
- Framework is Melitz (2003), but without international trade. In addition:
  - Firms produce using both labour and capital
  - Firms face "distortions" in the output market and in the capital market, so that their revenue net of variable costs are:

$$\pi_i = (1 - \tau_i^y) p_i y_i - w L_i - (1 + \tau_i^k) R K_i$$

- Distortions (often called wedges) are firm i specific!
- Possible to have 3 distortions (adding τ_i^ℓ), but one is redundant

Misallocation: A Note on Productivity

- Production function:

$$y_i = A_i K_i^{\alpha} L_i^{1-\alpha}$$

- We typically think of A_i as the firm's productivity (TFP). But Foster, Haltiwanger, and Syverson (2009) point out what we measure is "revenue productivity" (TFPR):

$$TFPR_i = A_i p_i$$

- Reason: we don't (typically) measure quantities in output data. Even if we did, very difficult to compare across products of varying quality, never mind across industries

- Going forward, generally good to differentiate between "quantity TFP" (TFPQ) and "revenue TFP" (TFPR)

Misallocation: A Note on Productivity cont.

- Production function:

$$y_i = A_i K_i^{\alpha} L_i^{1-\alpha}$$

- Similarly, marginal products of labour and capital can be defined in physical or revenue terms:

$$MPK_i = \frac{\partial y_i}{\partial K_i}, \qquad MRPK_i = \frac{\partial p_i y_i}{\partial K_i}$$

$$MPL_i = \frac{\partial y_i}{\partial L_i}, \qquad MRPL_i = \frac{\partial p_i y_i}{\partial L_i}$$

- Note that we don't observe these objects, but the firm's problem tells us something about what they should be equal to in equilibrium when the firm behaves optimally

Misallocation: The Firm's Problem

- CES demand $y_i = (p_i/P)^{-\sigma} Y$, and firms choose p_i, K_i, L_i to maximise π_i
- Price is a constant markup over marginal cost (+ sales taxes), scaled by the wedges:

$$p_i = \frac{\sigma}{\sigma - 1} \cdot \frac{1}{1 - \tau_i^y} \cdot \frac{1}{A_i} \left(\frac{R(1 + \tau_i^k)}{\alpha} \right)^{\alpha} \left(\frac{w}{1 - \alpha} \right)^{1-\alpha}$$

- Capital-labour ratio and marginal revenue products from firm optimality:

$$\frac{K_i}{L_i} = \frac{\alpha}{1 - \alpha} \frac{w}{R(1 + \tau_i^k)}, \qquad MRPL_i = \frac{w}{1 - \tau_i^y}, \qquad MRPK_i = \frac{R(1 + \tau_i^k)}{1 - \tau_i^y}$$

Misallocation: The Firm's Problem 2

- Note (whiteboard):

$$MRPL_i = (1-\alpha) \times TFPR_i \times \left(\frac{K_i}{L_i}\right)^{\alpha}$$

and similarly:

$$MRPK_i = \alpha \times TFPR_i \times \left(\frac{L_i}{K_i}\right)^{1-\alpha}$$

Misallocation: The Firm's Problem 3

- Consider the term $(MPRK_i)^{\alpha}(MPRL_i)^{1-\alpha}$ (board):

$$(MPRK_i)^{\alpha}(MPRL_i)^{1-\alpha} = TFPR_i \times \alpha^{1-\alpha}(1-\alpha)^{\alpha}$$

- So then, with distortions:

$$TFPR_i = \frac{1}{\alpha^{1-\alpha}(1-\alpha)^{\alpha}}(MPRK_i)^{\alpha}(MPRL_i)^{1-\alpha} = \left(\frac{R(1+\tau_i^k)}{1-\tau_i^y}\right)^{\alpha} \times \left(\frac{w}{1-\tau_i^y}\right)^{1-\alpha}$$

Misallocation: The Firm's Problem 4

- Cleaning up the algebra:

$$TFPR_i = \frac{R^{\alpha} w^{1-\alpha}}{\alpha^{1-\alpha}(1-\alpha)^{\alpha}} \times \frac{(1+\tau_i^k)^{\alpha}}{1-\tau_i^y}$$

- Absent wedges, we see that in equilibrium $TFPR_i$ is equalized across firms

Misallocation: Allocations

- In equilibrium, firms' after-tax MRPL and MRPK are equalized
- In the case where $\tau_i^k = \tau_i^y = 0$ for all i, we can see all firms charge identical markups $\frac{\sigma}{\sigma-1}$, and relative allocations are un-distorted
  - Note that the markups still cause some inefficiency (firms produce too little relative to socially optimal)!
- When τ_i differs across firms, allocations become distorted
  - To first order: τ_i^k distorts K_i/L_i, and τ_i^y distorts quantities
  - This causes dispersion in TFPR: some firms have higher or lower measured revenue TFP
  - Presence of TFPR dispersion is taken as a sign of misallocation

Misallocation: Firm-Level Wedges

- Can be shown with some algebra that:

$$1 + \tau_i^k = \frac{\alpha}{1-\alpha}\frac{wL_i}{RK_i}$$

$$1 - \tau_i^y = \frac{\sigma}{\sigma-1}\frac{wL_i}{(1-\alpha)p_i y_i}$$

$$A_i = \frac{(p_i y_i)^{\frac{\sigma}{\sigma-1}}}{K_i^{\alpha} L_i^{1-\alpha}}$$

- Exercise requires σ, α. In principle, α can be backed out from the data, but σ is calibrated

Misallocation: Aggregation

- Why does this matter? Exercise for homework: defining $L = \sum_i L_i$, $K = \sum_i K_i$, and $Y = \left(\sum_i y_i^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$ can be shown that if we define aggregate TFP such that

$$Y = TFP \times K^{\alpha} L^{1-\alpha}$$

then we have

$$TFP = \left[\sum_i \left(A_i \times \frac{\overline{TFPR}}{TFPR_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}}$$

where $\overline{TFPR}$ is (something like) average TFPR [the expression itself is quite nasty]

  - Or equivalently, $\overline{TFPR}$ the product $(\overline{MRPK})^{\alpha}(\overline{MRPL})^{1-\alpha}$ and some constants, where $\overline{MRPK}$ is the average MPRK

Misallocation: Aggregation, Special Cases

- If there is no dispersion in TFPR, then aggregate TFP is just a CES aggregate of firm TFP:

$$TFP = \left[\sum_i (A_i)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}$$

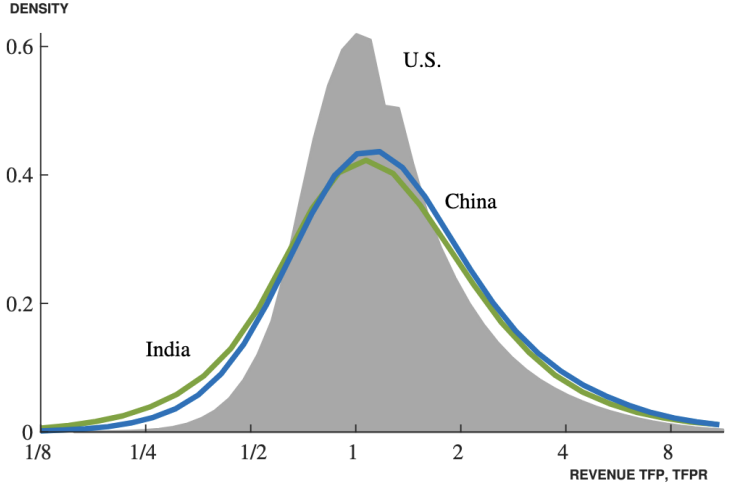
- If $TFPQ_i$ and $TFPR_i$ are jointly log-normal, then:

$$\log TFP = \frac{1}{\sigma - 1} \log \left(\sum_i (A_i)^{\sigma-1} \right) - \frac{\sigma}{2} \text{Var}(\log TFPR_i)$$

  - Here, TFPR dispersion shows up directly in the calculation for aggregate TFP

Misallocation: Application 1

- Hsieh and Klenow (2009) take data from US, India, and China and apply their framework



Misallocation: Application 2

- Previous figure implies that China + India's significant TFPR dispersion leaves room for productivity improvements via moving labour or capital from one firm to another!
  - Also true for the US, but less so.

- What would happen if there was no misallocation? Hsieh and Klenow (2009):

  "We freely admit this exercise **heroically** makes no allowance for measurement error or model misspecification. Such errors could lead us to overstate room for efficiency gains from better alloca- tion. With these caveats firmly in mind, Table IV provides percent TFP gains in each country from fully equalizing TFPR across plants in each industry."

Misallocation: Application 3

- Findings: under strong assumptions, misallocation *dramatically* reduces output across the three countries. The extent of misallocation is declining in China, but increasing in India from 1987-1994 and the US from 1977-1997.

TABLE IV
TFP GAINS FROM EQUALIZING TFPR WITHIN INDUSTRIES

China	1998	2001	2005
%	115.1	95.8	86.6
India	1987	1991	1994
%	100.4	102.1	127.5
United States	1977	1987	1997
%	36.1	30.7	42.9

Notes. Entries are $100(Y_{\text{efficient}}/Y - 1)$ where $Y/Y_{\text{efficient}} = \prod_{s=1}^{S}[\sum_{i=1}^{M_s}(\frac{A_{si}}{A_s}\frac{\overline{\text{TFPR}_s}}{\text{TFPR}_{si}})^{\sigma-1}]^{\theta_s/(\sigma-1)}$ and $\text{TFPR}_{si} \equiv \frac{P_{si}Y_{si}}{K_{si}^{\alpha_s}(w_{si}L_{si})^{1-\alpha_s}}$.

Misallocation: Application 4

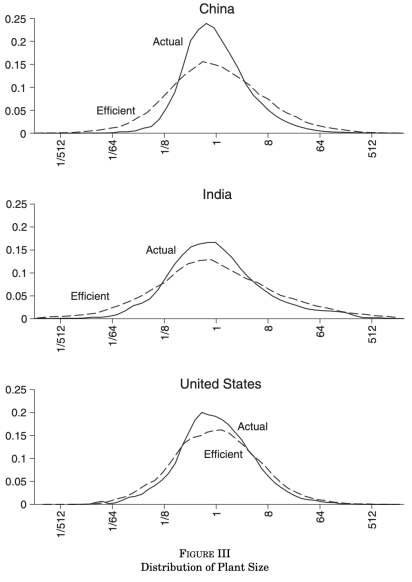


FIGURE III
Distribution of Plant Size

- Note: moving to the efficient benchmark does *not* make all plants the same TFPQ (and thus the same size)!
- There are still productivity differences across plants, and actually the plant size distribution widens, not shrinks

Misallocation to Variable Markups

- Via CES demand assumption, Hsieh and Klenow assume a constant markup $\frac{\sigma}{\sigma-1}$ for all plants
  - Variable markups potential source of misallocation? Hsieh and Klenow rule this out:

    > "Melitz and Ottaviano (2008) analyze the case of linear demand, under which the elasticity of demand is falling (and the markup increasing) with size... If linear demand applied everywhere, then TFPR should increase with size in the United States, too."

- Next: look at a canonical model of variable markups

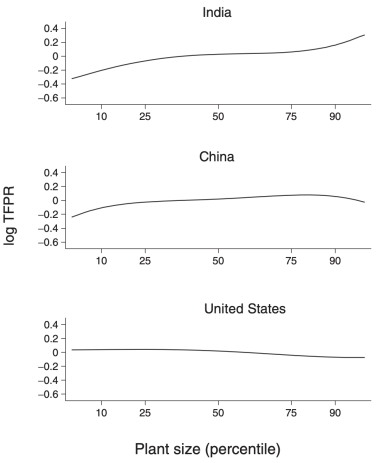


FIGURE VI
TFPR and Size

Variable Markups: Motivation

- Implausible to think that firms all charge the same markups. Generates strange predictions:
  - Complete passthrough: 1% rise in marginal cost increases prices by 1%
  - No pro-competitive effects: tougher competition doesn't impact markups
- In data, markups vary with size and productivity, and pass-through of cost shocks (e.g. exchange rates, tariffs) is incomplete

Variable Markups: Framework

- Going to follow Atkeson Burstein (2008), which focuses on strategic interactions between firms
  - Many alternatives to this, many of which change the curvature of demand
  - E.g., Kimball aggregator
- Idea: firms compete in one of many markets (think: industries, or product categories)
  - Within a market, firms internalize the effect of changes of their price on the price index
  - Across markets, firms do *not* internalize their impacts on their competitors
  - E.g., Ford competes directly against GM, but only indirectly against Apple

Variable Markups: Framework cont.

- Extend CES demand to *nested* CES. Let i denote firms, and j denote sectors, then:

$$U = \left(\int_j q_j^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}, \qquad q_j = \left(\sum_{i \in j} q_{ij}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

- Note: there is a finite number of firms in each market j, but an infinite number of markets
  - Can relax the latter assumption and just say firms don't internalize their impact on prices outside of their own sector
  - We assume $\sigma > \rho > 1$: products more substitutable within-sector than across sectors

Variable Markups: Demand

- Firm-level demand can be expressed:

$$q_{ij} = p_{ij}^{-\sigma} P_j^{\sigma-\rho} A_j$$

where A_j summarizes sectoral demand

- Thus: holding fixed P_j, elasticity of demand is just σ, as before. However, firms know that changes in their price changes P

- Result: firms' perceived demand elasticity is:

$$\epsilon_{ij} = \left[\frac{1 - s_{ij}}{\sigma} + \frac{s_{ij}}{\rho} \right]^{-1}$$

where s_i is their sales share in market j

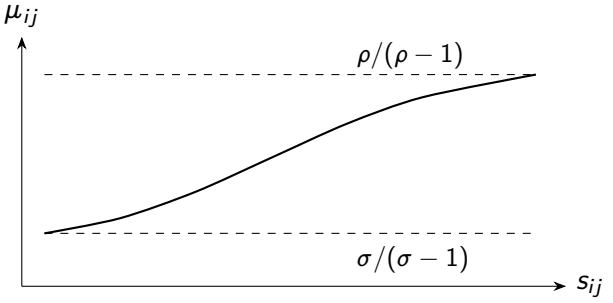
Variable Markups: Elasticity of Demand

$$\epsilon_{ij} = \left[\frac{1 - s_{ij}}{\sigma} + \frac{s_{ij}}{\rho} \right]^{-1}$$

- Two special cases:
  - $s_{ij} \to 1$: firm internalizes sectoral price effect, faces less elastic residual demand, and $\epsilon_{ij} \to \rho$
  - $s_{ij} \to 0$: firm behaves approximately like a monopolistically competitive firm, $\epsilon_{ij} \to \sigma$

Variable Markups: Markups

- As in standard CES, the markup is $\mu_{ij} = \frac{\epsilon_{ij}}{\epsilon_{ij}-1}$



- Productivity (cost) advantage $\rightarrow$ lower price, but that expands market share which offsets price reduction via higher markups

Variable Markups: Computation

- Computing markups given productivities φ_{ij} straightforward:

  1. Guess market shares s_{ij}
  2. Plug market shares into demand elasticities and markups
  3. Given $\{\varphi_{ij}\}$, compute p_{ij}, q_{ij}, P_j
  4. Re-compute market shares and return to step 2.
  5. Repeat until convergence

Estimating Markups: Motivation

- Atkeson Burstein: markups vary across firms. How large is this dispersion in the data?
  - Need $\mu_i = p_i/\lambda_i$, but marginal cost λ_i is not observed
  - Atkeson Burstein choose elasticities somewhat ad hoc. To properly estimate markups, need to estimate these parameters

- **Demand approach** (Berry, Levinsohn, and Pakes 1995): estimate demand, assume a model of conduct, back out the markup from the pricing first-order condition
  - Needs product-level prices and quantities, one market at a time

- **Production approach** (Hall 1988; De Loecker and Warzynski 2012): back out marginal cost from the firm's cost minimization problem
  - Needs only revenue and input expenditures, no model of demand or conduct

Production Approach: Setup

- Firm i produces $q_i = q_i(\varphi_i, n_i, k_i)$
- Timing assumption: n_i adjusts freely within the period, k_i is fixed at the start of the period (adjustment costs)
- Given k_i, the firm minimizes the cost of producing q_i, where λ_i is marginal cost
  - Note: we think of K_i as fixed here!

$$\mathcal{L} = w_i n_i + \lambda_i \left(Q - q(\varphi_i, n_i, k_i)\right)$$

- First-order condition with respect to n_i:

$$w = \lambda_i \frac{\partial q_i}{\partial n_i}$$

Production Approach: The Markup

- Multiply the first-order condition by n_i/q_i to get the output elasticity of n_i:

$$\theta_i^n \equiv \frac{\partial q_i}{\partial n_i}\frac{n_i}{q_i} = \frac{1}{\lambda_i}\frac{w_i n_i}{q_i}$$

- Using $\mu_i = p_i/\lambda_i$:

$$\mu_i = \theta_i^n \cdot \frac{p_i q_i}{w_i n_i}$$

- Markup = elasticity / expenditure share of n_i (share observed, elasticity estimated)
  - No assumptions on demand, structure of competition, or returns to scale
  - Perfect competition: $\mu_i = 1$, so elasticity = revenue share
  - Note: we are assuming n_i fully flexible!

Estimating Output Elasticities

- Production function in logs:

$$q_{it} = f(n_{it}, k_{it}; \beta) + \log \varphi_{it} + u_{it}$$

  where φ_{it} is productivity and u_{it} is a shock unseen when choosing inputs

- Cobb-Douglas gives $\theta^n_{it} = \beta_n$

- Problems with OLS:
  - Simultaneity: firms see ω_{it} before choosing v_{it}, so the two are correlated
  - Data: usually only revenue is observed, not prices and quantities

Estimating Output Elasticities: Simultaneity

- Suppose production function in logs is

$$\log q_{it} = \beta_n \log n_{it} + \beta_k \log k_{it} + \log \varphi_{it} + u_{it}$$

- OLS estimates of β_n, β_k are generally biased because:

$$\mathbb{E}[n_{it}\varphi_{it}], \mathbb{E}[k_{it}\varphi_{it}] \neq 0$$

Estimating Output Elasticities: Data

- Most firm level datasets observe revenue (sales), not p and q separately
- Output elasticity θ here is physical output elasticity, not revenue
- With variable markups, responses of $p_i q_i$ to demand or productivity shocks can come from changes in p [markups] or q

Estimating Output Elasticities: Second Stage

- Given β, back out productivity: $\varphi_{it}(\beta) = \hat{\varphi}_{it} - f(v_{it}, k_{it}; \beta)$
- Markov process $\omega_{it} = g(\omega_{i,t-1}) + \xi_{it}$. Regress $\omega_{it}(\beta)$ on a polynomial in $\omega_{i,t-1}(\beta)$ to get the innovation $\xi_{it}(\beta)$
- Capital is predetermined and lagged inputs are uncorrelated with ξ_{it}, so use GMM on

$$E\left[\xi_{it}(\beta)\begin{pmatrix} k_{it} \\ v_{i,t-1} \end{pmatrix}\right] = 0$$

- Then $\hat{\theta}^n_{it} = \partial f / \partial v$ at $\hat{\beta}$, and $\hat{\mu}_{it} = \hat{\theta}^V_{it} P_{it} Q_{it} / (P^V_{it} V_{it})$

Estimating Markups: Implementation

- De Loecker, Eeckhout, and Unger (2020): financial statements of US public firms (Compustat), 1955–2016
  - Sales $\to P_{it}Q_{it}$, cost of goods sold (COGS) $\to P_{it}^{V}V_{it}$, gross PP&E $\to K_{it}$
  - Overhead (SG&A) is treated as a fixed cost
- Output elasticities θ_{st}: Cobb-Douglas, estimated by 2-digit sector, varying over time
  - Robustness: cost shares (valid if all inputs are flexible and CRS)
- Firm-level markup: $\hat{\mu}_{it} = \hat{\theta}_{st} \times \text{Sales}_{it}/\text{COGS}_{it}$
- Aggregate markup: $\mu_t = \sum_i m_{it}\mu_{it}$, with m_{it} the firm's share of total sales

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